NATIONAL INSTITUTE OF TRANSPORT(NIT)

DEPARTMENT OF TRANSPORT AND MARINE ENGENERING

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SHIP PROPULSION:

1:CONCEPT OF SHIP PROPULSION

- PROPULSION: Refers to a system that provides a force that enable an object to move forward. i.e, ship, car, plane, jet or rocket .
- SHIP: Refers to a large marine vessel that is used to transport people and their cargoes from one place to another through water as a means of transport.
- SHIP PROPULSION: Refers to a ship system which provides a force that enable it to move forward through water.

OR

In a more broader terms ship propulsion can be described as follows; Ship propulsion is the method used to move a ship through water by generating thrust to overcome resistance.

- BASIC PRINCPLE OF SHIP PROPULSION:

1. The ship propulsion pushes water backward using the energy generated from the main engine(with the aid of propellers or water jets)
2. By newton third law of motion the water provides a reaction which in return pushes the ship forward.
3. And the generated force is called thrust which enable the ship to overcome resistance and move forward.

2:MAIN COMPONENTS OF SHIP PROPULSION

1. POWER PRODUCER (Prime mover)

- This is the main power producer (engine). It has various types which are;

1.1 Diesel engine: this is a type of engine which operates on compression ignition where air is highly compressed causing its temperature to rise and fuel is injected which self-ignites without spark.

>Basic principle ;

1. Works on compression ignition
2. No spark plug used
3. Fuel ignites due to high temperature of compressed air

1.2 Gas turbines; this is type of engine that uses combustion produces power by using high temperature and high velocity gases to rotate a turbine

>Basic principle

1. works on Brayton cycle

2.Air is compressed, fuel is burned, and the hot gasses expand through a turbine to produce power

1.3 Steam turbine; this is rotary heat engine that converts the thermal energy of steam through turbine blades

>basic principle

1. works on the Rankine cycle.

2. High- pressure, high-temperature steam expands through turbine nozzles and blades.

3. the change in momentum of steam cause the turbine rotor to rotate.

1.4 Electric motor; Is the type of an engine that converts electrical energy into mechanical energy by electromagnetic action.

>basic principle

1. works on electromagnetic induction.

2. A current-carrying conductor placed in a magnetic field experiences a force causing rotation

2. TRANSMISSION SYSTEM (Power transfer)

2.1 shafting; this is the system of shafts and associated components used to transmit power and rotational motion from the main engine or motor to the propeller in ship. That includes, thrust shaft, intermediate shaft, tail shaft, couplings, and bearings

2.2 gearbox; this is used to match engine speed to propeller speed and may also provide reverse drive.

3. PROPULSOR (Converts power into thrust)

3.1. Propeller; this is a rotating mechanical device with blades that converts rotational power from the engine into thrust, moving a ship forward by pushing water backwards
3.2. Water jet; is the ship propulsion system in which water is drawn into the vessel and expelled at high velocity through a nozzle producing thrust that moves the ship forward.
3.3. Pod drive; this is propulsion system in which an electric motor is housted inside a submerged, steerable pod that can rotate through 360 °, providing both propulsion and steering without the need for a conventional rudder.

3: TYPES OF PROPULSION:

3.1 Propeller propulsion;

Mode of action; a propeller pushes water backward to create a thrust that pushes a ship forward.

>Uses;

+used in cargo ships

+used in tankers

+used in container ships

ADVANTAGES;

- ✓ Its simple and efficient
- ✓ Its reliable
- ✓ Low coast

3.2 Water jet propulsion;

Mode of action; water is sucked in and expelled at a high speed creating a thrust to move a ship forward.

>Uses;

+Used in fast ferries

+Used in naval vessels

ADVANTAGES;

- ✓ It produces high speed
- ✓ It is suitable for shallow draft operation
- ✓ It gives good maneuverability

3.3 Diesel engine propulsion

Mode of action; fuel is burned by the heat generated by the compressed gas (no spark used) and so pressure generated pushes the pistons to generate power.

>uses;

+Used in cargo ships

+Used in container ships

ADVANTAGES;

- ✓ Produces high thermal efficiency
- ✓ Low fuel consumption
- ✓ It is reliable and durable

3.4 Electric/diesel-electric propulsion

Mode of action; the engine generates electricity which turns the electric motors which drives the propeller.

>uses;

+used in cruise ships

+used in icebreakers

+used in offshore vessels

ADVANTAGES;

- ✓ Provides high efficiency
- ✓ Produces excellent maneuverability
- ✓ It provides low noise and vibration
- ✓ Flexible machinery layout
- ✓ Reduces fuel consumption
- ✓ Lower maintenance
- ✓ It is environment friendly
- ✓ Its highly reliable

3.5 Podded propulsion;

Mode of action; the electrical motor is connected inside a rotating pod which enable it to rotate in 360°

>Uses;

+used in high speed vessels

ADVANTAGES;

- ✓ It provides excellent maneuverability
- ✓ No rudder needed

4:COMPARISON OF TYPES OF PROPULSION:

SYSTEM	SPEED	EFFICIENCY	USE
Propeller	medium	high	Cargo ships
Water jet	high	medium	Fast vessels
Diesel	Medium	Very high	Merchant ships
Electric	Medium	High	Cruise ships

5: IMPORTANCE OF SHIP PROPULSION

- ✓ It helps to determine speed
- ✓ Affects fuel consumption
- ✓ Influences maneuverability
- ✓ It has impact on operational cost

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